In summary:

4.8.4 Theorem Matrices $\mathbf{G}$ and $\mathbf{H}$ are generating and parity check matrices respectively for a linear (n,k)-code C iff

- ★ The rows of $\mathbf{G}$ are linearly independent,
- ★ The columns of $\mathbf{H}$ are linearly independent,
- ★ $\mathbf{G}$ is $k \times n$, and $\mathbf{H}$ is $n \times (n-k)$, and
- ★ $\mathbf{G}\,\mathbf{H} = \mathbf{0}$.

Then the rows of $\mathbf{G}$ are a basis for C, and the columns of $\mathbf{H}$ are a basis for $C^\perp$. The relationship between $\mathbf{G}$ and $\mathbf{H}$ is summarized in diagram 4.7.1.

(4.7.1)

$$\begin{tikzcd}
\text{Code} & \text{Generating Matrix} & \text{Parity Check Matrix} \\
\overset{(n,k)}{C} & \overset{k\times n}{\mathbf{G}_C} \arrow[r, bend left, "4.6.13"] & \overset{n\times(n-k)}{\mathbf{H}_C} \arrow[dl, "\mathsf{T}"] \\
\overset{(n,n-k)}{C^\perp} & \underset{(n-k)\times n}{\mathbf{H}_C^{\mathsf{T}} = \mathbf{G}_{C^\perp}} \arrow[r, "4.6.13"'] & \underset{n\times k}{\mathbf{G}_C^{\mathsf{T}} = \mathbf{H}_{C^\perp}} \arrow[ul, "\mathsf{T}"']
\end{tikzcd}$$

If $G = (I_k \; X)$ is a generating matrix for an (n,k)-code C then $H = \begin{pmatrix} X \\ I_{n-k} \end{pmatrix}$ is a parity check matrix for C.

The next theorem shows an important use of the parity check matrix. Recall that by definition of $\mathbf{H}$, the column space of $\mathbf{H}$ is $C^{\perp}$.

4.8.1 Theorem Let C be a linear code with parity check matrix $\mathbf{H}$. Then $C = \{v \in \mathbb{F}_2^n \mid v\mathbf{H} = \mathbf{0}\}$.

Proof Clearly $v\mathbf{H} = \mathbf{0}$ iff v is perpendicular to each column of $\mathbf{H}$ iff v is perpendicular to the column space of $\mathbf{H}$ (see Lemma 4.6.4) iff $v \in (C^{\perp})^{\perp} = C$. □

In particular: *If we receive w and $w\mathbf{H} \neq \mathbf{0}$, then error(s) must have occurred.*

<u>Exercise</u> The matrix $H = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is a parity check matrix for a linear code C.

Which of the following words are codewords of C?

(a) 01110

(b) 10011

(c) 10100

For a linear (n, k)-code, the matrix equation $x\mathbf{H} = \mathbf{0}$, where $x = (x_1, x_2, \ldots, x_n)$, is a shorthand way of writing a set of $n - k$ simultaneous linear equations in the variables $x_1, x_2, \ldots, x_n$. If the jth column of $\mathbf{H}$ has entries of 1 in positions $m_1, m_2, \ldots m_j$, then the bit-sum

$$x_{m_1} + x_{m_2} + \cdots + x_{m_j} = 0.$$

In other words, the binary word x satisfies $x\mathbf{H} = \mathbf{0}$ iff the sums (obtained by ordinary addition) of certain of its bits (as determined by $\mathbf{H}$) are even (have even parity). The $n - k$ linear equations are thus referred to as *parity check equations* of the code, and $\mathbf{H}$ is called the *parity check matrix*.

4.9. Distance of a Linear Code

4.9.1 Theorem Let **H** be a parity check matrix for a linear code C. Then δ is equal to the smallest number (> 0) of rows of **H** that are linearly dependent.

Proof Let S be a non-empty set of rows of **H** of minimal cardinality such that S is linearly dependent. Let $s = |S|$. We must show $s = \delta$.

$\boxed{\delta \leq s}$ Since S is dependent, there exists T with $\emptyset \neq T \subseteq S$ such that the sum of elements of T is $\mathbf{0}$. Minimality of S implies $S = T$, so S sums to $\mathbf{0}$.

Form the word b of length n whose ith bit is 1 iff row i of **H** is in S. Thus $\|b\| = |S| = s > 0$. And $b\mathbf{H}$ is exactly the sum of words in S. So $b\mathbf{H} = \mathbf{0}$ and $b \in C \setminus \{\mathbf{0}\}$. By Theorem 4.2.1, $\delta \leq \|b\| = s$.

$\boxed{s \leq \delta}$ By definition of δ there exists $c \in C \setminus \{\mathbf{0}\}$ with $\|c\| = \delta$. Let R be the set of rows of **H** corresponding to the positions where the bits of c are 1. Thus $|R| = \|c\| = \delta$.

Now $c \in C$ implies $\mathbf{0} = c\mathbf{H} =$ the sum of the elements of R. So R is a linearly dependent set of size δ. Minimality of S implies $s \leq \delta$. $\square$

<u>Exercise</u> The following matrices are parity check matrices for codes C, D and E, respectively.

$$H_C = \begin{pmatrix} 1&0&1 \\ 0&1&1 \\ 1&0&1 \\ 1&1&0 \\ 0&0&1 \end{pmatrix} \qquad H_D = \begin{pmatrix} 1&1 \\ 1&0 \\ 0&1 \\ 1&0 \\ 0&0 \end{pmatrix} \qquad H_E = \begin{pmatrix} 1&1&0 \\ 0&1&1 \\ 1&0&0 \\ 0&1&0 \\ 0&0&1 \end{pmatrix}$$

Determine the distance of each of these codes.

4.10. Cosets

4.10.1 Definition If C is a linear code of length n and $u \in \mathbb{F}_2^n$, then the *coset of C containing u* is the set $\{c + u \mid c \in C\}$. We denote this coset by $C + u$ (or $u + C$).

The coset $(C + e)$ containing e is exactly the set of all possible received words if error pattern e occurred.

Example Consider $C = \{0000, 0101, 1010, 1111\}$.

$C + 1100 =$

$C + 0101 =$

4.10.3 Theorem [Properties of Cosets] Let C be a linear (n, k)-code. Let u, $v \in \mathbb{F}_2^n$. Then:

(1) $u \in C + u$.

(2) $u + v \in C \iff u \in C + v \iff C + u = C + v$.

(3) Every word in $\mathbb{F}_2^n$ is contained in exactly one coset of C: that is, either $C + u = C + v$ or $(C + u) \cap (C + v) = \emptyset$.

(4) $|C + u| = |C|$.

(5) There are 2^{n-k} different cosets of C. Each coset contains 2^k words.

(6) The code C is itself one of the cosets.

Proof

(1) Because C is linear, it contains the zero vector. So $0 + u \in C + u$ so $u \in C + u$.

(2) We first prove that $u + v \in C \implies u \in C + v$. If $u + v \in C$, then $u + v = c$ for some $c \in C$. Thus, $u = c + v$ and so $u \in C + v$.

We now prove that $u \in C + v \implies C + u = C + v$. If $u \in C + v$, then $u = c + v$ for some $c \in C$. Consider any element $c_1 + u \in C + u$. We have $c_1 + u = c_1 + c + v \in C + v$. So $C + u \subseteq C + v$. Similarly, if $c_2 + v$ is any element of $C + v$, then $c_2 + v = c_2 + c + u \in C + u$. So $C + v \subseteq C + u$ and we have $C + u = C + v$.

We now prove that $C + u = C + v \implies u + v \in C$. If $C + u = C + v$, then $u \in C + u = C + v$. Thus, $u = c + v$ for some $c \in C$ and so $u + v = c \in C$.

(3) We show that the relation $\sim$ on $\mathbb{F}_2^n$ given by $u \sim v \iff u + v \in C$ is an equivalence relation. Reflexive: $u + u = 0 \in C$ so $\sim$ is reflexive. Symmetric: $u + v \in C$ implies $v + u \in C$, so $\sim$ is symmetric. Transitive: $u + v \in C$ and $v + w \in C$ implies $u + w = u + v + v + w \in C$, so $\sim$ is transitive. By (2) the equivalence classes of this relation are the cosets, and this gives us the result.

(4) Consider the function $f : C \to C + u$ given by $f(c) = c + u$ for all $c \in C$. If $c_1 + u = c_2 + u$, then $c_1 = c_2$ so f is injective, and f is clearly surjective so $|C + u| = |C|$.

(5) The cosets partition $\mathbb{F}_2^n$ by (3). By (4), each coset has size $|C + u| = |C| = 2^k$, so there must be exactly 2^{n-k} of them.

(6) C is the coset $C + 0$. $\square$

Exercise Consider again the code $C = \{0000, 0101, 1010, 1111\}$.

(a) How many cosets does C have?

(b) List all the cosets of the code C.

4.11. IMLD for Linear Codes

One of the fundamental goals of coding theory is to design codes which permit easy and rapid decoding of a received word. We describe such a method for IMLD for a linear code, using the parity check matrix and cosets of the code.

Recall (Theorem 4.8.1) that for a linear code C, $v \in C$ if and only if $vH = 0$, where H is a parity check matrix for C.

4.11.1 Lemma Let C be a linear code with parity check matrix $\mathbf{H}$. Then $u\,\mathbf{H} = v\,\mathbf{H} \iff C + u = C + v$.

Proof $u\,\mathbf{H} = v\,\mathbf{H} \iff (u+v)\,\mathbf{H} = \mathbf{0} \iff u + v \in C \overset{4.10.3}{\iff} C + u = C + v.$ □

Let C be a linear code. Assume the codeword $c \in C$ is transmitted and the word w is received, with an error pattern $e = c + w$. Clearly, $w + e = c \in C$, so $C + w = C + e$. That is, *the error pattern e is in the same coset as the received word w* by Theorem 4.10.3 (2).

4.11.2 Definition Any word of minimal weight in a coset is called a *coset leader* for that coset.

There may be several words of equal smallest weight, so the coset leader is not necessarily unique.

4.11.3 Algorithm [Decoding using cosets] Suppose C is a linear code and the word w is received. If the coset $C + w$ has a unique coset leader e, conclude that $c = w + e$ was the word sent.

4.11.4 Example Let $C = \{0000, \quad 1011, \quad 0101, \quad 1110\}$ as in Example 4.10.4. The cosets of C are:

Coset 1	Coset 2	Coset 3	Coset 4
0000	1000	0100	0010
1011	0011	1111	1001
0101	1101	0001	0111
1110	0110	1010	1100

Exercise For each of the following received words, determine the most likely sent codeword.

(a) 0111

(b) 1110

(c) 1111

4.11.5 Definition Let C be a linear (n, k)-code with parity check matrix H. For any word $w \in \mathbb{F}_2^n$, define the *syndrome* of w to be $w\mathrm{H}$. This is a word in $\mathbb{F}_2^{n-k}$.

The point of this is that *words from the same coset have the same syndrome:* Words u and v are in the same coset iff $u + v \in C$ (Theorem 4.10.3(2)) iff

$$\mathbf{0} = (u+v)\mathrm{H} = u\mathrm{H} + v\mathrm{H} \qquad \Longleftrightarrow \qquad u\mathrm{H} = v\mathrm{H}.$$

Since words in the same coset have the same syndrome and words in different cosets have different syndromes, we can identify a coset by the syndrome of any word in the coset. Thus for an (n, k)-code, the 2^{n-k} words of length $n-k$ each occur as the syndrome of exactly one of the 2^{n-k} cosets.

A Standard Decoding Array (SDA) for a linear code C is a table that matches each syndrome with its coset leader(s).

One way to construct an SDA is to list all the cosets of C and find the coset leaders. Then find a parity check matrix for C and calculate the syndrome for each coset leader.

4.11.7 Theorem Let C be a linear code of length n, H be a parity check matrix for C and w, $e \in \mathbb{F}_2^n$. Then

★ $w\mathrm{H} = \mathbf{0}$ iff w is a codeword in C.

★ $w\mathrm{H} = e\mathrm{H}$ iff w and e lie in the same coset of C.

★ w is corrected using the coset leader e from the coset containing w (if unique).

★ If e is the error pattern for a received word w then $e\mathrm{H}$ is the sum of the rows of H corresponding to the positions where errors occurred in transmission.

IMLD for a linear code C using an SDA

Given an SDA and the corresponding parity check matrix H:

For each received word w

- calculate wH
- find the syndrome wH in the SDA
- if the syndrome has a unique coset leader e, then conclude c = w + e was sent.

  if the syndrome does not have a unique coset leader, then request retransmission.

4.11.10 Example Let $C = \{00000,\ 10100,\ 01011,\ 11111\}$ (this is linear). Construct an SDA for C.

Solution: First we calculate the cosets of C:

$$\begin{aligned}
C &= \{00000,\ 10100,\ 01011,\ 11111\}\\
10000 + C &= \{10000,\ 00100,\ 11011,\ 01111\}\\
01000 + C &= \{01000,\ 11100,\ 00011,\ 10111\}\\
00010 + C &= \{00010,\ 10110,\ 01001,\ 11101\}\\
00001 + C &= \{00001,\ 10101,\ 01010,\ 11110\}\\
11000 + C &= \{11000,\ 01100,\ 10011,\ 00111\}\\
10010 + C &= \{10010,\ 00110,\ 11001,\ 01101\}\\
10001 + C &= \{10001,\ 00101,\ 11010,\ 01110\}
\end{aligned}$$

Next we find a parity check matrix $\mathbf{H}$. We can use generating matrix

$$\mathbf{G} = \left(\begin{array}{cc|ccc} 1 & 0 & 1 & 0 & 0\\ 0 & 1 & 0 & 1 & 1 \end{array}\right) = (\mathbf{I}\ \ \mathbf{X}).$$

$$\mathbf{H} = \begin{pmatrix} \mathbf{X} \\ \mathbf{I} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0\\ 0 & 1 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1 \end{pmatrix}.$$

Coset Leader(s)	Syndrome
00000	000
10000, 00100	100
01000	011
00010	010
00001	001
11000, 01100	111
10010, 00110	110
10001, 00101	101

Exercise Use the syndrome decoding array for the code $C = \{00000, 10100, 01011, 11111\}$ given above to apply IMLD to the following received words.

(a) 01001

(b) 11000

(c) 10101

Exercise

Construct an SDA for the linear code $C = \{0000, 1011, 0101, 1110\}$.

Coset 1	Coset 2	Coset 3	Coset 4
0000	1000	0100	0010
1011	0011	1111	1001
0101	1101	0001	0111
1110	0110	1010	1100

Consider the linear code $C = \{000, 111\}$.

(a) Determine the values of n, k, δ.

(b) Find a generating matrix G and a parity check matrix H.

(c) How many cosets does C have?

4.11.12 Example Let C be the code with generating matrix $\mathbf{G}$ and parity check matrix $\mathbf{H}$ where

$$\mathbf{G} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} \qquad \mathbf{H} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(a) Determine the parameters (n, k, δ) of this code.

(b) How many cosets does this code have?

The two codes just discussed are Hamming codes. Decoding of Hamming codes is very simple.

- ★ Receive word w. Calculate its syndrome $s = w\,\mathbf{H}$.
- ★ If $s = \mathbf{0}$, w requires no correction.
- ★ Otherwise, the syndrome will be the jth row of $\mathbf{H}$, for some unique j. Correct bit j of w.

4.12. Encoding and Decoding Summary

We summarize the information transmission process for a linear (n, k)-code. The possible messages are the 2^k words of length k, that is, all the vectors in the vector space $\mathbb{F}_2^k$. The codewords are 2^k of the possible 2^n words of length n, and the set of all these codewords, the code, is a subspace of the vector space $\mathbb{F}_2^n$. The received words are all the possible words of length n. Let G and H be a generating matrix and parity check matrix, respectively.

★ Message word m of length k

★ Codeword $c = m\mathrm{G}$ of length n

★ Received word w of length n

★ Compute $w\mathrm{H}$

★ Most likely error pattern e is the coset leader of the coset with syndrome $w\mathrm{H}$.

★ Most likely codeword is $c' = w + e$

4.13. Reliability of IMLD for Linear Codes

Recall that $Q_C(c_0)$ is the probability that if codeword c_0 is sent, then the received word is corrected to c_0.

Also Q_C is the average of $Q_C(c_0)$ over all codewords $c_0 \in C$,

so $Q_C = \frac{1}{|C|} \sum_{c_0 \in C} Q_C(c_0)$.

$$\boxed{\text{For linear codes } Q_C(c_0) = Q_C(c_1) \ \forall\, c_0, c_1 \in C.}$$

Recall that any received word w is decoded to $w+e$ where e is the unique coset leader for coset $C+w$.

(If $C+w$ does not have a unique coset leader, then w cannot be decoded and we request retransmission.)

Thus, w is correctly decoded if and only if the error pattern e is a unique coset leader.

So $$Q_C = \sum_{e \in \mathcal{E}} p^{n-\|e\|}(1-p)^{\|e\|}$$

where $\mathcal{E}$ is the set of unique coset leaders of C.

Recall the code and SDA from Example 4.11.10.

Let $C = \{00000,\ 10100,\ 01011,\ 11111\}$

Coset Leader(s)	Syndrome
00000	000
10000, 00100	100
01000	011
00010	010
00001	001
11000, 01100	111
10010, 00110	110
10001, 00101	101

Exercise

Suppose codewords of C are transmitted over a BSC with reliability $p = 0.99$. Determine Q_C.

In-class assignment - next week in your applied class.

Work in groups of 2 or 3.

Hand in one answer sheet per group.

Open book and you can get help from staff.

You'll be given a generating matrix G for a linear code.

Answer questions like - what are n, k?
- is C systematic?
- find parity check matrix H
- what is distance of C?

Then you need to construct an SDA <u>without listing all the cosets</u>, and use the SDA to find most likely codewords (and message words) corresponding to a string of received words.

Finally you will be asked about the reliability of IMLD for this code.

4.14. Generalising to larger fields: q-ary codes

4.14.1 Definition Let $n \geq k$ be positive integers. A *(q-ary block) code* consists of:

- ★ A set $M \subseteq \mathbb{F}_q^k$ of *message words* together with
- ★ An injective function called the *encoding function* $f : M \to \mathbb{F}_q^n$.

We write $C = f(M)$, the image of f, and (abusing terminology) call C the *code*. Elements in C are called *codewords*. The *length* of the code is n, and the *message length* or *dimension* of the code is k.

A code C is an (n, k) q-ary code if it has codewords of length n and $|C| = q^k$.

Note that the distance between two words x and y in $\mathbb{F}_q^n$ is the number of places in which they differ. The relationships between distance and error detection and correction are the same for q-ary codes as for binary codes. Note that when codeword c is transmitted and error pattern e occurs, then the received word is $w = c + e$, but that for non-binary codes, this means $e = w - c$ and $c = w - e$.

The concept of a generating matrix, parity check matrix, dual code, and syndrome decoding all generalise naturally from binary to q-ary codes. Note however, the following important difference in determining a parity check matrix from a generating matrix in standard form. Let

$$\mathrm{G} = (\mathrm{I}_k \;\; \mathrm{X})$$

where X is some $k \times (n-k)$ matrix, be a generating matrix for an (n, k) q-ary code. Then

$$\mathrm{H} = \begin{pmatrix} -\mathrm{X} \\ \mathrm{I}_{n-k} \end{pmatrix}$$

is a parity check matrix for this code. By block multiplication, we see that

$$\mathrm{G\,H} = (\mathrm{I}_k \;\; \mathrm{X}) \begin{pmatrix} -\mathrm{X} \\ \mathrm{I}_{n-k} \end{pmatrix} = -\mathrm{I}_k\,\mathrm{X} + \mathrm{X}\,\mathrm{I}_{n-k} = 0_{k \times (n-k)}.$$

Exercise: Consider the ternary code with generating matrix

$$\begin{pmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 1 & 2 \end{pmatrix}$$

(a) Determine n and k for this code.

(b) Determine the codeword corresponding to the message word 21.

(c) Determine a parity check matrix for this code, and use it to find the distance δ.

(d) Construct an SDA for this code and use it to find the most likely message word for the received word 2211.